

RailBEV: A Vision-Based Bird's-Eye-View Representation Network for 3-D Railway Object Detection

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Abstract—Three-dimensional railway object detection is vital to the environmental perception of intelligent railway transportation systems. However, owing to the complexity of railway environments, critical issues such as the detection of multiscale and distant objects remain to be overcome. This study proposes a vision-based architecture called RailBEV, which introduces bird's-eye-view (BEV) representation within 3-D railway object detection. RailBEV is mainly divided into two parallel parts: the depth supervision view transformation (DSVT) and long-range supervision (LRS) modules. Specifically, the DSVT module designed a novel depth supervision method to transform the front view (FV) images into the BEV plane. The LRS module leverages an additional 3-D detection head to guide a 2-D image backbone to learn extra 3-D object information and enhance the capability of long-range object detection. Experimental results demonstrate that the proposed RailBEV exhibits a mean average precision (mAP) of 38.6%, which is 4.4% higher than the baseline 3-D object detection network BEVDepth. Moreover, the results show that RailBEV effectively supports long-range detection and identifies multiscale and short-range objects. All these indicate that RailBEV has the potential to be readily applicable in 3-D railway object detection.

Index Terms—Bird's-eye-view (BEV) representation, deep learning, depth supervision, object detection, railway safety.

I. INTRODUCTION

WITH the rapid development of intelligent railway transportation, train operations are evolving from manual control to autonomous driving systems [1], [2]. For the safety of self-driving trains, accurate environmental perception is essential [3], [4]. The perception system helps detect obstacles during train operation, thereby preventing collisions and improving efficiency. The detection of 3-D railway objects is a fundamental element in the perception of railway environments.

A. Background

To improve the safety of railways, extensive researchers have concentrated on detecting objects in railway

environments and achieved remarkable progress. Railway object detection is currently divided into two approaches: 2-D and 3-D detection. Recent studies in 2-D detection have improved the accuracy of small object detection in railway scenarios through methods such as multiscale feature fusion [5] and global feature integration [6]. While these models achieve high accuracy in 2-D railway object detection, they are limited to predicting object locations without providing 3-D spatial information.

Beyond 2-D image-based approaches, several point cloud-based methods have been developed for 3-D railway object detection. Researchers leverage methods such as end-to-end 3-D detection [7], Transformer attention mechanisms [8], and 4-D millimeter-wave radar point cloud clustering [9]. While these point cloud-based methods have demonstrated promising results in 3-D railway object detection, they suffer from low computational efficiency, rendering them unsuitable for real-time deployment. Furthermore, constrained by light detection and ranging (LiDAR) performance limitations, these methods exhibit an effective detection range below 200 m. Considering the above issues, it is vital to develop a suitable method for long-range and lightweight 3-D railway object detection.

In recent years, vision-based 3-D object detection methods have achieved significant progress due to their lower computational cost. Some methods [10], [11] focus on the front view (FV) and utilize 2-D object detection networks to predict 3-D objects, which often fail to predict the location of objects accurately, particularly at long ranges. On the other hand, bird's eye view (BEV) in object detection networks [12], [13], [14] has gained popularity owing to its excellent spatial representation capability, making it particularly advantageous for 3-D object detection, as shown in Fig. 1. A detection model based on BEV includes a backbone, a view transformation module (VTM), and a 3-D object detection head. The core of these three components is the VTM, whose primary function is projecting FV image features onto the BEV plane through depth estimation for each pixel and corresponding pixels between FV features and BEV features. Nevertheless, current VTMs are designed for short-range detection (within 60 m) on open roads. They cannot satisfy the requirements of long-distance and differently sized object detection in railways.

Motivated by the success of BEV-based networks [15], [16], [17], [18], this study innovatively introduced BEV into railway object detection and designed a VTM for railways to enhance BEV semantic features and improve detection accuracy.

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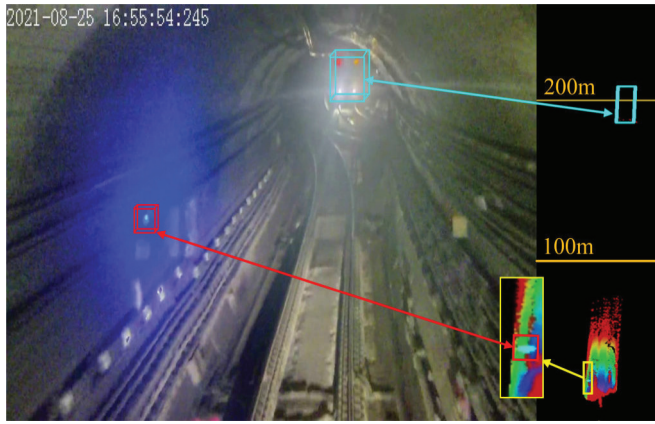


Fig. 1. Example of 3-D railway bounding boxes. The left image presents the FV, while the right image presents the BEV perspective. The red box denotes a railway signal light with dimensions $0.15 \times 0.2 \times 0.2$ m (ordered by length, width, and height, respectively). The blue box denotes a train, whose size is $22 \times 2.8 \times 3.8$ m. Additionally, yellow horizontal lines serve as distance markers at 100- and 200-m intervals within the BEV image.

This research introduces a novel vision-based network called RailBEV, which is designed for long-range 3-D detection of railway objects. Here, based on operational criteria, long range is defined as distances exceeding 120 m. As illustrated in Fig. 2, RailBEV has two parallel modules: the depth supervision view transformation (DSVT) module and the long-range supervision (LRS) module.

The DSVT module is proposed to transform the FV image into a BEV representation, allowing for the extraction of spatial depth information at the pixel level and facilitating 3-D depth estimation. To improve the accuracy of depth estimation, the depth map generated by RailDepth [19] is utilized as ground truth to supervise the depth estimation process.

Additionally, the LRS module is specifically designed to improve long-range 3-D object detection. As the object distance increases, the accuracy of 3-D position prediction tends to degrade. To address this challenge, an additional detection head is introduced to refine the feature extraction process, enabling the model to learn and represent long-range 3-D object information more effectively.

B. Contributions

The main contributions of this study are summarized as follows.

- 1) The BEV-representation network was integrated into the vision-based 3-D railway object detection domain. A DSVT module was proposed to augment the spatial depth understanding during the view transform process, thus improving object detection ability.
- 2) An LRS module was designed to detect distant objects. Adjusting the standard FV 2-D image backbone to fit the BEV framework is a pivotal step. Supervision was implemented through a heatmap head to capture the object depth and lateral positioning.
- 3) RailBEV, a vision-based 3-D railway object detection network, was formulated to ensure optimal precision for 3-D detection across diverse railway scenarios.

Compared to point-cloud-based approaches, vision-based 3-D object detection networks exhibited higher accuracy in detecting distant and multiscale objects.

C. Article's Organization

The organization of this article is shown as follows. Section II provides a summary of relevant works in railway object detection and BEV-based 3-D object detection. Section III details the methodology of the RailBEV network. Section IV analyzes the experimental findings, and Section V provides a conclusion with summary and suggestions for future work.

II. RELATED WORK

Recently, increasing scholars have focused on enhancing railway safety through object detection [20]. This section reviews railway object detection methods based on 2-D images and 3-D point clouds. Then, recent advancements in BEV-representation object detection networks are introduced.

A. Railway Object Detection

Existing railway object detection methods are primarily classified into image-based 2-D object detection and point cloud-based 3-D object detection. For 2-D railway object detection, Lian et al. [5] proposed an intensive multiscale fusion network to enhance the accuracy of 2-D small obstacle detection in railway scenarios by integrating semantic features. As a result, the detection accuracy in railway scenarios improved to a mean average precision (mAP) of 75.16%, with a reduced processing time of 10.5 ms/image. Furthermore, leveraging the Transformer architecture, Zhao et al. [6] proposed AENet, a train object detection model that enhances global feature understanding, enabling the precise detection of 2-D small objects. This approach achieves an mAP of 95.3% in railway scenarios, with a processing time of 40.7 ms/image. Fioretti et al. [21] devised a method to reconstruct the trajectories of trains that exhibited considerable potential for obstacle detection. Ye et al. [22] introduced a two-stage detection method DFF-Net. The second-stage integrated semantic information from the initial level via a differential feature fusion module and achieved remarkable accuracy. The method attains an mAP of 79.9% in train scenarios, with a processing time of 18.5 ms/image. He et al. [23] proposed a flexible network FE-YOLO, comprising a CRBlock, an attention framework, and an SPF module. A novel aggregation module was developed to detect small irregular obstacles. This method achieves an mAP of 76.1% in train obstacle detection, with a processing time of 10.0 ms/image. Despite existing 2-D railway object detection methods having achieved remarkable results in bounding box detection, they fail to accurately predict the distance and spatial information of obstacles, which are crucial for intelligent railway perception and avoiding collisions.

Beyond 2-D image-based approaches, several point cloud-based methods have been developed for railway 3-D object detection. Yuan et al. [7] designed an end-to-end railway 3-D object detection network, utilizing a Transformer-based framework to transform 3-D point clouds into sparse 2-D

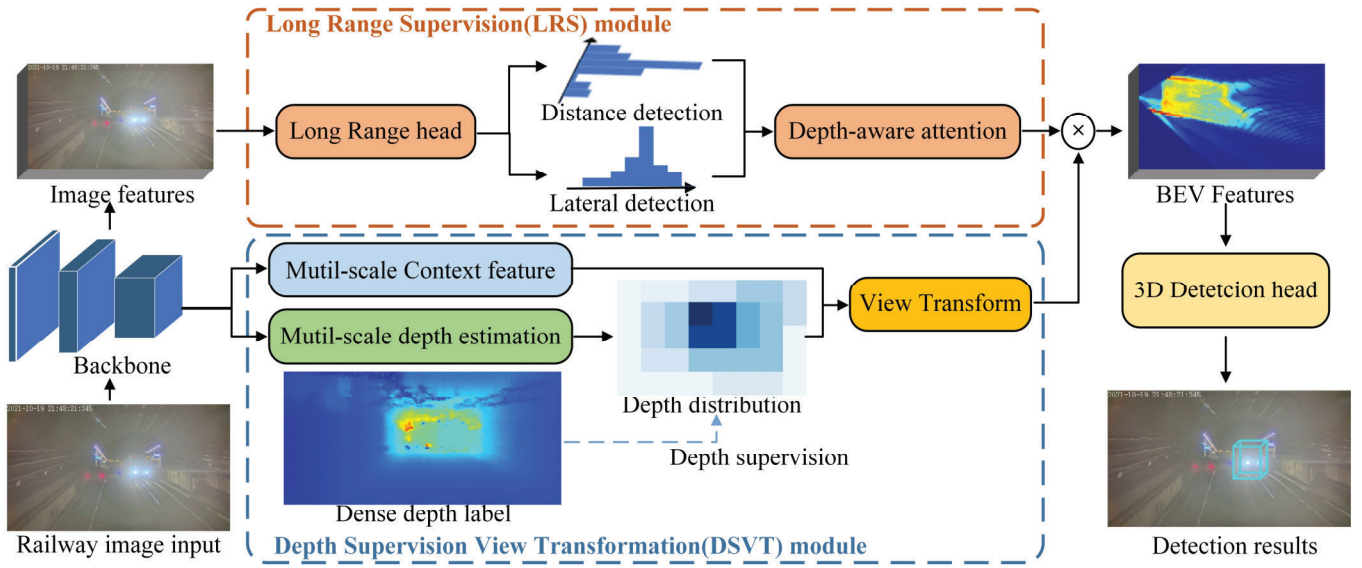


Fig. 2. Framework of RailBEV. First, the ResNet-50 is utilized to obtain image features. Subsequently, the DSVT module is designed to convert the FV feature into a BEV representation, enhancing the spatial depth information during view transformation. Additionally, the LRS module guides the features to adapt 3-D object detection by directly introducing 3-D knowledge into 2-D image Backbone. Finally, an anchor-free detection head is designed to obtain the 3-D bounding box of obstacles.

features, followed by a traditional 2-D detection head to estimate object locations in 3-D space. The final accuracy achieved was 91.0% mAP. Lian et al. [8] proposed a method for railway track detection using 3-D point clouds, where the point cloud data are first converted into a 2-D depth map to reduce computational complexity. The proposed method achieved a 3-D mAP of 87.5%, with a per-frame processing speed of 130 ms. Wang et al. [9] introduced a 4-D radar-based method for train 3-D object detection in adverse weather conditions, applying density-based spatial clustering of applications with noise (DBSCAN) clustering to refine the obstacle detection range and accurately localize the 3-D center position of the trains. The accuracy of 3-D railway object detection is generally higher than 2-D object detection, and it can predict the 3-D location information of obstacles. However, most existing 3-D detection approaches are optimized for detecting nearby objects and struggle with long-range detection. Also, the computational speed needs to be improved. Therefore, there is a need to investigate real-time 3-D object detection methods optimized for railway environments.

B. Vision-Based 3-D Object Detection

Although cameras cannot offer accurate depth measurements like point clouds, the rich semantic information in images enables vision-based 3-D object detection methods to acquire remarkable accuracy [24]. Early approaches in this domain primarily relied on predicting 3-D objects from 2-D image features. FCOS3D [25] employs a straightforward structure to predict the 2-D center of a 3-D object using 2-D semantic features. It estimates the 3-D object's position alongside its corresponding depth information, demonstrating the feasibility of leveraging 2-D features for 3-D tasks. PGD [26] simplifies the 3-D object detection task by transforming it into depth estimation for 2-D objects. By incorporating spatial

geometric relationships among multiple objects, PGD significantly improves depth estimation accuracy, leading to more precise 3-D object detection. DETR3D [15] is the first model to achieve end-to-end 3-D object detection without requiring additional post-processing. By leveraging the Transformer mechanism for 3-D environment modeling and designing a dedicated 3-D object detection head, DETR3D significantly improves the accuracy of detecting long-range objects. PETR [27] introduced 3-D position-aware representations to replace traditional detection heads, mapping 2-D image features to 3-D spatial points and enhancing detection precision. PETRv2 [28] incorporates a temporal modeling module that aggregates data from multiple historical frames to capture temporal features. This enhancement significantly improves the accuracy of downstream tasks, such as object tracking and velocity prediction.

In contrast to the methods for directly detecting 3-D objects from 2-D features, a novel methodology has recently emerged. Recently, LSS [29] introduced a method that converts FV image features into BEV representations. First, the LSS method predicted the depth value in the FV image, utilizing camera calibration data to determine the 3-D spatial position of each pixel. Based on the LSS method, BEVDet [16] employed the anchor-free head from CenterPoint [30] to predict 3-D objects using BEV features generated via view transformation. It can achieve comparable accuracy and reduce computational demands. The subsequent work BEVDet4D [31] introduces temporal information with processed multiple keyframes simultaneously. Concurrently, BEVFormer [17] utilized the deformable attention mechanism to generate BEV features. However, the accuracy of depth estimation in the VTM remains low, leading to errors in 3-D object detection. To address this problem, BEVDepth [18] generates depth image based on point clouds to guide the VTM. This

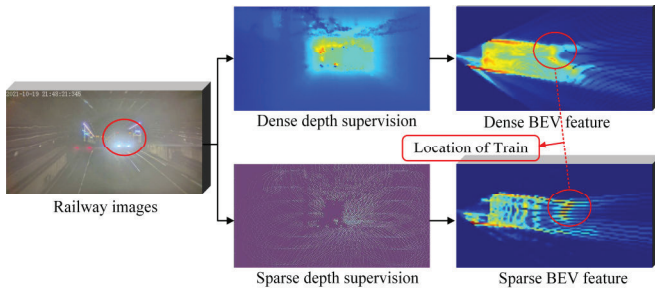


Fig. 3. Comparison between dense depth supervision module and sparse depth supervision module. FV features are transformed into BEV features using sparse or dense depth supervision methods, as depicted in the middle image. Notably, the red circles represent the position of the train object.

approach showed that accurate depth predictions can substantially improve view transformation performance and improve the detection results. Furthermore, SA-BEV [32] employed semantic information to exclude redundant background pixels and predict only the foreground object depth, which can mitigate the issue of object information being overwhelmed by a substantial amount of background data. PolarBEVDet [33] introduces a BEV perspective transformation based on the polar coordinate system, ensuring uniform distribution of image information and preserving feature consistency during transformation. Full self-driving (FSD)-BEV [34] designs a foreground self-distillation module to learn semantic features from point cloud-based 3-D object detection, enabling the 2-D feature extractor to capture 3-D spatial representations during feature extraction.

From these studies, it is evident that optimizing the VTM performance remains significant for BEV object detection. Depth supervision has demonstrated excellent performance in enhancing depth estimation and VTMs. Existing depth supervision methods such as BEVDepth [19], BEVStereo [35], and Fb-BEV [36] generate ground-truth depth images using sparse LiDAR point clouds. However, considering the restricted detection radius of point clouds, the ground-truth depth image tends to be overly sparse, as shown in Fig. 3. A sparse depth image results in diminished detection accuracy for long-range objects and multisize objects. Thus, a novel depth supervision architecture is introduced, utilizing dense depth images as the ground truth generated by a depth completion network and employing a multiscale depth supervision network to facilitate view transformation.

III. METHODOLOGY

The purpose of this work is to achieve precise 3-D railway object detection. For this purpose, this study proposes RailBEV, a BEV-based network that is explicitly designed for 3-D railway object detection. RailBEV comprises two parallel modules: the DSVT and the LRS. Section III-A details the DSVT module. First, a dense depth image is generated as the ground truth. Then, a multiscale depth estimation framework with a mixed-pooling loss function was designed. In parallel, the LRS module was used to improve the accuracy of detecting long-distance objects. An additional 3-D detection head was proposed in the LRS module to learn the knowledge of distant

objects, as outlined in Section III-B. The predicted heatmaps of the objects are input as weights into the BEV features.

A. DSVT Module

In this section, a DSVT module is proposed to supervise the depth estimation during view transformation. First, a dense depth ground truth was generated using the depth completion network RailDepth [19]. Subsequently, a novel multiscale depth supervision network was designed to fuse the depth information. Finally, a mixed-pooling loss is introduced to supervise the depth prediction results.

1) *Generated Dense Depth Image*: The premise of the DSVT module is to generate ground truth of depth images. However, all existing depth supervision methods obtain depth images from sparse point cloud data [18], [35], [36], as shown in the sparse depth supervision of Fig. 3. A 2.5-D image coordinate depth ground truth $P_i^{\text{img}}(u, v, d)$ can be derived utilizing the transformation matrix from LiDAR to camera and intrinsic camera parameters. However, these depth images from LiDAR have two problems: 1) the sparse point cloud leads to the absence of depth information and 2) short detection distance. Currently, the detection distance of LiDAR is less than 100 m, which is not sufficient for railway object detection.

RailDepthv2, an enhanced railway depth completion tool, was employed to overcome the limitations. RailDepthv2 is built on RailDepth [19] and is capable of reconstructing dense depth images from sparse LiDAR and RGB data. The original RailDepth employed the hourglass backbone [37], introducing a fusion module along with a unique pooling guidance loss. Crucially, as a self-supervised network, it leverages only the original point clouds and images, which is advantageous for extensive depth ground-truth acquisition.

Nevertheless, the real-time application of RailDepth requires a balance between computation speed and accuracy, which results in the precision of depth completion being insufficient for the ground truth. Hence, RailDepthv2 is proposed with two improvements: 1) a more extensive backbone, transitioning from Hourglass-18 to Hourglass-34 and a more expansive feature pyramid network (FPN) [38] for multiscale features, which can enhance spatial and semantic extraction and 2) a higher resolution feature map by reducing the downsampling rate of image features. Higher resolution can diminish the blurriness of object edge depths, thereby providing more accurate depth values for objects.

The final dense depth ground truth and corresponding BEV features are shown in the upper part of Fig. 3. The generated dense depth supervision ground truth provides additional semantic details and 3-D information, facilitating the transformation of FV features into BEV features. This significantly enhances the density of BEV features and captures more object details. For instance, as highlighted by the red circles in Fig. 3, the train object exhibits blurred contours in sparse BEV features, while the dense BEV features generated by the DSVT module provide more object information. This improvement boosts the accuracy of subsequent 3-D object position predictions.

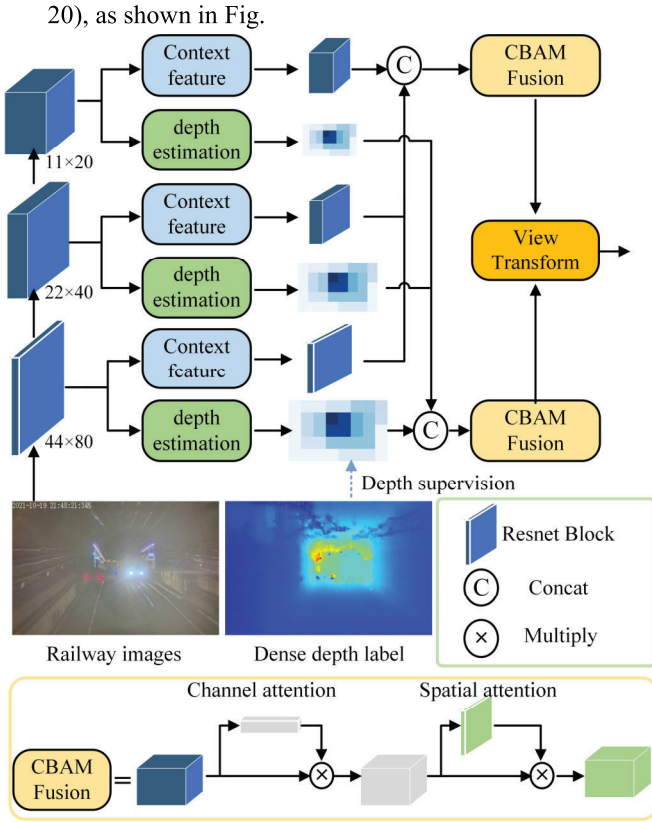


Fig. 4. Architecture of multiscale depth supervision network. Depth estimation is performed on multiscale features individually and the resulting features are integrated by CBAM fusion.

Compared with the original sparse depth labels, denser BEV features are obtained with the DSVT module. Simultaneously, the module provides accurate predictions for depth estimation and object boundaries. For instance, the red dashed rectangle shows the train features with distinct boundaries, whereas the corresponding features in original methods are ambiguous. This indicates that the DSVT module effectively highlights the train's position, resulting in accurate detection.

2) *Multiscale Depth Supervision Network*: This study explored the transformation of image features to the BEV representation by calculating the depth of each pixel and leveraging camera calibration information for 3-D spatial positioning. As is evident, depth estimation accuracy is essential for BEV features. However, foundational detectors such as LSS [29] and BEVDet [16] derive depth estimation supervision solely from detection loss. Considering the challenges of monocular depth estimation, this method is insufficient for supervising the VTM.

Therefore, the intermediate depth estimation is proposed to be supervised. Considering the railway operation complexities and object size variations, a multiscale depth supervision network was designed to ensure accurate depth estimation. Compared with existing depth supervision methods, depth estimation was conducted on three scales (44×80 , 22×40 , and 11×20), as shown in Fig. 4.

For large objects such as trains, the importance of accurate distance outweighs detailed boundaries. Thus, low-resolution

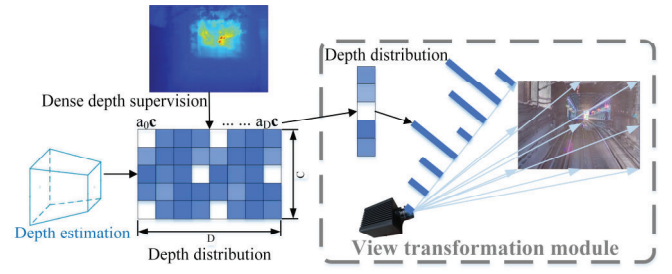


Fig. 5. Diagram of VTM.

(11×20) features are utilized to allocate channels and capture the accurate distance. Conversely, small objects, such as signal lights, require high-resolution (44×80) features to prevent losing depth information. Notably, the channel numbers for all scales are halved during view transformation to minimize additional computations.

Specifically, the depth supervision module is divided into two components at each scale: the context feature layer and the depth estimation module. The context feature layer is a 3×3 kernel convolution to change the number of feature channels. The depth estimation module comprises a residual block and a dilated convolution layer. A residual block is used for feature extraction and the dilated convolution is used to expand the receptive field of feature maps.

After obtaining multiscale depth information and context features, all of them are unsampled to 44×80 resolution and concatenated to obtain the last depth estimation D_{pred} and context feature F_{context} . Inspired by the convolutional block attention module (CBAM) [39], a feature fusion module is proposed for adjusting features. By employing channel attention, varying weights are allocated to individual channels to process diverse channel information, while spatial attention emphasizes the localization of significant features. The module then calculates the attention weights across channels and spatial features and multiplies it with the original feature map to adaptively adjust features.

As shown in Fig. 5, after obtaining the depth map via the multiscale depth supervision network, the depth of each pixel is transformed into a depth distribution range map $D = d_0 + \Delta, \dots, d_0 + |D|\Delta$. The depth distribution is represented as a $|D|$ -dimensional vector, indicating that the probability that the pixel's depth falls within $|D|$ discrete depth ranges. Here, d_0 denotes the base depth, which is set to 0 in this study, and Δ represents a discrete depth range unit. For an image $X \in R^{3 \times H \times W}$, let p be a pixel at coordinates (u, v) with feature c . The probability that the pixel p has a depth of $d_0 + d * \Delta$ is denoted by a_d . To associate the 2-D pixel p with the 3-D point $p_d(u, v, d) \in R^3 | d \in D$, this study employs a projection method to construct 3-D spatial grid features. The feature c_d at point p_d is computed as shown in the following equation:

$$c_d = a_d * c. \quad (1)$$

3) *Mixed-Pooling Loss*: In the previous depth supervision loss function, a 352×640 resolution depth label is down-sampled by a factor of 8 to produce a tensor $GT_{\text{depth_raw}}$ with dimensions $44 \times 80 \times 64$. The following step uses a

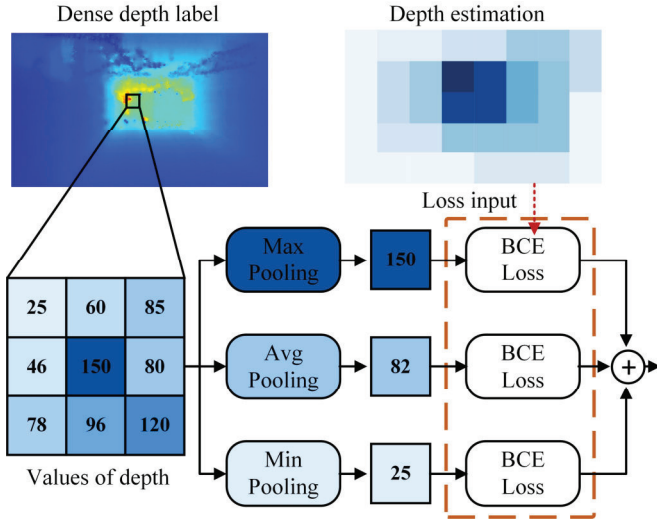


Fig. 6. Mixed-pooling loss calculation process.

minimum pooling layer to determine the minimum depth value per pixel, generating the final depth ground truth GT_{depth} , whose resolution is $44 \times 80 \times 1$.

However, the min-pooling layer results in significant information loss from long-range depths. Although this is advantageous for short-range detection, it is unsuitable for railway object detection. To address this, a mixed-pooling loss function is proposed, which separately applies min, max, and average pooling to GT_{depth_raw} to acquire three ground-truth depth labels GT_{depth_min} , GT_{depth_max} , and GT_{depth_avg} , as shown in Fig. 6. The loss is computed for each of the three ground-truth values. Finally, the three losses are summed to create a comprehensive depth supervision loss function, which enables the depth supervision network to learn over a spectrum of distances rather than focusing solely on the closest ones. The mixed-pooling loss adopts binary cross entropy (BCE) [40] and the loss function is shown as follows:

$$L_{depth} = \frac{\sum_{i=1}^3 l_{BCE}(D_{pred}, GT_i)}{W \cdot H} \quad (2)$$

where L_{depth} is the mixed pooling loss. Parameters W and H are the width and height of final ground truth of depth, which means 80 and 44, respectively. l_{BCE} is the BCE loss function. GT_i represents the three ground-truth depth labels, GT_{depth_min} , GT_{depth_max} , and GT_{depth_avg} .

B. LRS Module

In this section, an LRS module is proposed to enhance the detection accuracy within a 100–260-m range. Initially, an additional LRS head was devised to predict both the distance and lateral positions of distant objects, guiding the 2-D image backbone to learn long-range information. Subsequently, depth and lateral position heatmaps are generated based on the long-range detection head. A depth-aware attention mechanism is proposed to integrate the predicted heatmap weights into the BEV features derived from the DSVT module.

1) *Long-Range Detection Head*: BEV-representation 3-D object detection networks predict 3-D bounding boxes based on BEV features. Therefore, the final 3-D object detection loss function is also applied to the BEV features, which is implicit with the image backbone in FV. However, general image backbone networks such as ResNet [41] and VoVNet [42], along with their pretrained models, are designed for 2-D railway tasks such as 2-D object detection and segmentation. It may not be ideal for 3-D railway object detection. Thus, it is imperative to directly supervise these feature backbones to guide them in learning 3-D knowledge, which is typically absent in 2-D recognition tasks.

Certain networks directly supervise the image backbone, such as BEVformerv2 [43]. All of them are based on extensive feature backbones like VoVNet-99, ResNet101, or Swin Transformer and directly predict the normal 3-D bounding boxes, including position, size, and category of objects. However, because of the real-time demands, ResNet-50 is adopted as the feature backbone, which inevitably leads to poor performance when directly predicting complete 3-D bounding boxes during the feature extraction phase. This makes guiding the backbone toward 3-D knowledge acquisition challenging. Furthermore, due to the large loss values, predicting full 3-D bounding boxes can result in unstable training or nonconvergence.

To overcome the problems, an LRS head is proposed, which focuses solely on predicting the distance and lateral position of 3-D objects. The LRS head simplifies the prediction task and enables the 2-D image backbone to assimilate 3-D object insights. As shown in Fig. 7, the LRS head uses features from 2-D backbone as inputs and processes them via a 3×3 kernel convolutional layer. Efficient channel attention (ECA) is employed to obtain one-hot vectors for each feature. Then, two detection heads based on convolution layers are utilized to predict depth and lateral position.

Considering the remarkable results of existing backbones in short-range detection, the 2-D feature extractor is supervised to learn knowledge of distant 3-D objects. To simulate railway scenarios, two heatmaps HT_{dis} and HT_{lat} are generated for distance and lateral position, which gradually decrease from the object's center. Referring to the heatmap strategy of CenterNet [44], the decay radius is set to 3. Like the depth supervision loss, the BCE loss is utilized to address the imbalance between positive and negative samples. The loss function of LRS module is shown as follows:

$$L_{LRS} = \frac{f_{BCE}(F_{dis} - HT_{dis})}{l_{depth}} + \frac{f_{BCE}(F_{lat} - HT_{lat})}{l_{lat}} \quad (3)$$

where L_{LRS} is the LRS loss. Parameters l_{dis} and l_{lat} refer to the length of one-hot vectors HT_{dis} and HT_{lat} . F_{dis} and F_{lat} are results of the LRS detection head. f_{BCE} is the BCE loss function.

2) *Depth-Aware Attention Architecture*: The LRS module guides the 2-D image backbone to acquire 3-D knowledge, and its predicted heatmap can also be utilized as weights for the BEV features to enhance the long-range detection. Considering the characteristics of the BEV representation, each pixel in BEV features can be interpreted as semantic information of the corresponding spatial position.

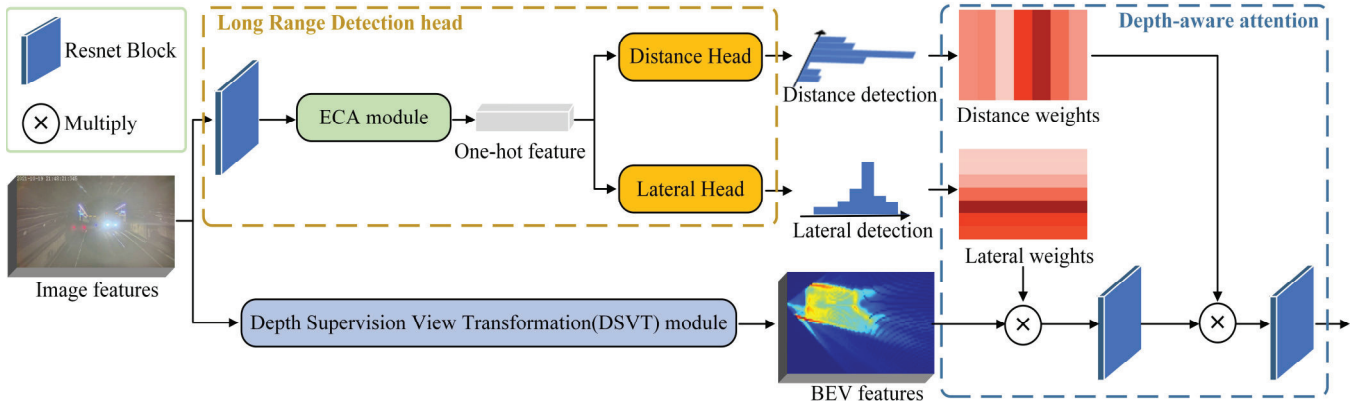


Fig. 7. Architecture of LRS module. A long-range detection head is designed to predict the location heatmap of objects from 2-D image features. The depth-aware attention converts the location heatmap into attention weights, which are integrated into the BEV features.

For example, a pixel at (i, j) in the BEV features can be associated with a spatial location that is i meters forward and j meters laterally from the coordinate origin. Hence, by integrating the heatmap as an attention weight into the BEV features, it enables the features to acquire additional distant object information. As shown in the LRS of Fig. 7, the distance attention W_{dis} and lateral attention W_{lat} are separately introduced into the BEV features to ensure that the attention is learned efficiently. The depth-aware attention function is shown as follows:

$$F_{\text{BEV}} = f_{\text{res}} \left(F_{\text{BEV}} + F_{\text{BEV}}^{(100,260)} \times W_{\text{dis}} \right) \quad (4)$$

$$F_{\text{BEV}} = f_{\text{res}} \left(F_{\text{BEV}} + F_{\text{BEV}}^{(-30,30)} \times W_{\text{lat}} \right) \quad (5)$$

where F_{BEV} is the BEV features. $F_{\text{BEV}}^{(100,260)}$ refers to the BEV feature values within a depth range of 100–260 m. Similarly, $F_{\text{BEV}}^{(-30,30)}$ refers to the BEV features values within a lateral range of -30 to 30 m. f_{res} is a residual block from ResNet. The above two functions are computed sequentially, with the depth attention calculated first, followed by the lateral attention.

C. RailBEV Architecture

For 3-D railway object detection, the RailBEV network was designed utilizing the DSVT module and the LRS module.

The structure of RailBEV comprises three components: image backbone, BEV view transformation, and 3-D object detection head. As depicted in Fig. 2, single-frame railway images are used as input. First, a ResNet50 network is used as a backbone network to extract the 2-D image features. The BEV VTM is composed of two parallel components: the DSVT module and the LRS module. The DSVT module facilitates the transformation from FV to the BEV plane. The LRS module is added to supervise the backbone to learn 3-D object knowledge and utilize the predicted heatmaps as weights for BEV features. Finally, a CenterPoint-based anchor-free 3-D detection head is used to obtain the results. An anchor-free 3-D detection head is designed based on CenterPoint, which relies on object heatmaps to determine the object position and size. Compared with the traditional anchor-based 3-D detection heads such as Second [45] and PointPillar

[46], anchor-free methods do not require predesigning anchor box sizes. The proposed 3-D detection head is suitable for multiscale object detection and offers greater robustness across different environments.

Furthermore, considering the balance between detection accuracy and model complexity, a cascade multiscale structure is used to gather more features for the depth supervision module. Concurrently, an FPN [38] is applied to detect long-range and multiple types of objects. The final output size of the 3-D detection head is

$$S_{\text{output}} = W \times H \times (n_c + n_{xyz} + n_{\text{whl}} + n_{\text{yaw}} + n_{\text{vel}}) \quad (6)$$

where S_{output} is the final output prediction result. W and H are the width and height of the last heatmaps, respectively, i.e., $W = H = 256$ in this study. n_c is equal to 4, representing the number of object categories. Similarly, n_{xyz} is equal to 3, representing the position of the object's center in 3-D space. n_{whl} is equal to 3, representing the size of objects. n_{yaw} is equal to 2, representing the rotation of an objects' 3-D bounding box. n_{vel} is equal to 2, representing the velocity of objects.

The loss function comprises four parts: depth supervision loss, LRS loss, category loss, and bounding box loss. For the category loss, a generalized focal loss is applied, and L_1 loss is utilized for bounding box loss. The formulation of the loss function is as follows:

$$\text{Loss} = L_{\text{depth}} + L_{\text{LRS}} + w_{\text{cls}} L_{\text{cls}} + w_{\text{box}} L_{\text{box}} \quad (7)$$

$$L_{\text{cls}} = \sum_i^{n_c} f_{\text{GFL}}(C_i - \hat{C}_i) \quad (8)$$

$$L_{\text{box}} = \frac{\sum_i^{n_{\text{head}}} \left(w_i \sum_j^{n_{\text{max}}} |x_{ij} - \hat{x}_{ij}| \right)}{n_{\text{max}}} \quad (9)$$

where L_{depth} , L_{LRS} , L_{cls} , and L_{box} stand for depth supervision loss, LRS loss, category loss, and bounding box loss, respectively. w_{box} is the weight of bounding box loss, which is equal to 0.25 in this study. For L_{cls} , f_{GFL} is the generalized focal loss function. C_i is the estimation confidence of category, and $\hat{C}_i$ is the confidence of the GT. f_{GFL} calculates the confidence loss for each category separately. For L_{box} , n_{max} is the number of objects predicted by the model, which equals 500 in this article. n_{head} is the number of detection heads mentioned

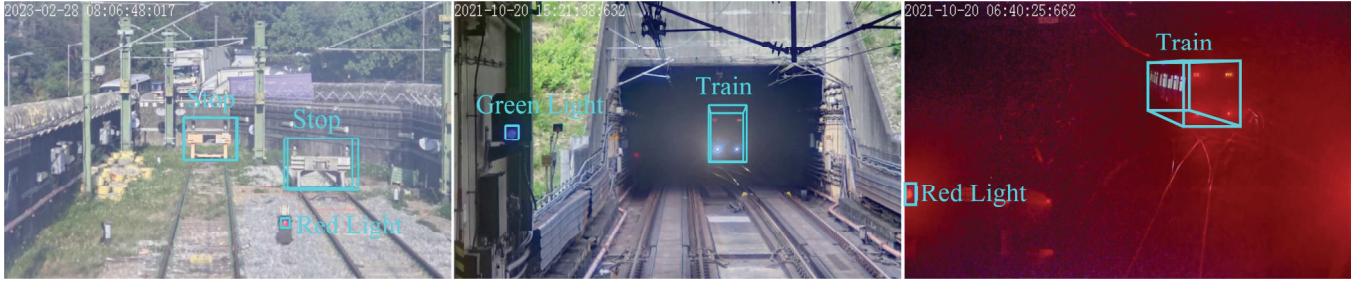


Fig. 8. Examples of railway dataset. The blue boxes refer to the ground truth of 3-D objects. The two images on the left represent daytime scenes, while the image on the far right represents a tunnel scene. These examples demonstrate that the dataset contains numerous multiscale objects and has challenging lighting conditions, which significantly increases the difficulty of 3-D object detection.

TABLE I
OBJECT CLASS LIST

Type	Train	Red Light	Green Light	Stop
training	17150	5372	670	7972
testing	3871	915	288	1330
ratio	4.4:1	5.8:1	2.3:1	5.9:1

above, including position, size, rotation, and velocity. w_i is the weight of the four detection heads. x_{ij} is the prediction value of each object of each detection head, and $\hat{x}_{ij}$ is the ground truth for x_{ij} .

IV. EXPERIMENTS

This section presents the experiments designed to assess the effectiveness of the proposed approach.

A. Dataset and Setup

To analyze the method quantitatively, this study collected large-scale calibrated video frames and LiDAR frames and established a 3-D railway object detection dataset. The dataset was collected from the Hong Kong Metro Tsuen Wan line and comprises synchronized images and LiDAR data. The image resolution is 1280×720 . The point cloud comes from a long-range LiDAR Livox tele15 with a detection range exceeding 300 m. The raw data were annotated following the Nuscenes dataset [47] format to develop the final 3-D object dataset. The Hong Kong Metro Tsuen Wan line is 16 km long and contains 16 stations. The dataset was collected on the entire route when the railway was operating, and typical objects were selected and annotated as shown in Fig. 8.

The railway dataset contains 22 878 finely labeled samples (18 608 for training and 4270 for testing). The training set and testing set are distributed in a ratio of approximately 4.3–1. The labeling categories include “Train,” “Red Light,” “Green Light,” and “Stop” (“Stop” stands for the marker at the end of rail track as shown in Fig. 8), which are listed in Table I. Notably, the ratio of training to testing sets for the bounding boxes of each category was maintained within a reasonable range.

To assess the accuracy of RailBEV, this study utilized standard evaluation metrics commonly applied in 3-D object

detection tasks from the Nuscenes dataset, including mAP, Nuscenes detection score (NDS), average translation error (ATE), average scale error (ASE), and average orientation error (AOE). The details of each evaluation metric are as follows.

1) *Mean Average Precision (mAP)*: The average precision (AP) metrics commonly used in 2-D object detection are adapted with slight modifications. Instead of using intersection over union (IOU), a match is defined based on the thresholding of the 2-D center distance d on the ground plane. The AP is computed across a range of matching thresholds $\mathbb{D} = \{0.5, 1, 2, 4\}$ m and for each class in set $\mathbb{C}$. Thus, $AP_{c,d}$ denotes the AP for class c at distance d

$$mAP = \frac{1}{|\mathbb{C}||\mathbb{D}|} \sum_{c \in \mathbb{C}} \sum_{d \in \mathbb{D}} AP_{c,d}. \quad (10)$$

2) *True Positive (TP) Metrics*: Beyond the mAP, the TP metrics are employed to assess methods. These TP metrics are computed based on a center distance of $d = 2m$ during matching and are formulated as positive scalars. The following TP errors are defined.

The ATE measures the Euclidean distance between the centers of predicted and true locations in 2-D, expressed in meters. The ASE quantifies the 3-D IOU discrepancy after aligning orientation and translation, calculated as $(1 - IOU)$. The AOE evaluates the smallest yaw angle error between ground truth and predictions

$$mTP = \frac{1}{|\mathbb{C}|} \sum_{c \in \mathbb{C}} TP_c. \quad (11)$$

3) *Nuscenes Detection Score (NDS)*: Given that mAP does not fully encompass all dimensions of 3-D object detection tasks, such as attribute estimation, the nuScenes dataset introduces a new metric. This metric, known as the NDS, avoids condensing various error types into a single scalar score, providing a more comprehensive assessment of detection performance

$$NDS = \frac{1}{10} \left[5 mAP + \sum_{mTP \in TP} (1 - \min(1, mTP)) + 2 \right] \quad (12)$$

where TP is the set of the three mean TP metrics.

The pytorch1.10 is used to conduct experiments, the networks are trained using four NVIDIA 4090 GPUs, and inference is performed on a NVIDIA 3090 GPU.

TABLE II
COMPARISONS WITH EXISTING OBJECT DETECTION NETWORKS

Method	Backbone	Temporal	mAP $\uparrow$	NDS $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	mAOE $\downarrow$	#Parameter	FPS
PETR [27]	VOVNet-99	✗	0.263	0.390	0.762	0.646	1.519	81663204	33.2
BEVDet [16]	ResNet50	✗	0.274	0.520	0.566	0.133	0.466	54648573	54.7
BEVFormer [17]	ResNet50	✓	0.285	0.409	0.691	0.640	1.530	33511905	32.5
BEVDet4D [31]	ResNet50	✓	0.293	0.529	0.562	0.108	0.499	55858893	17.1
PETRv2 [28]	VOVNet-99	✓	0.274	0.400	0.722	0.642	1.525	85107498	22.3
BEVDepth [18]	ResNet50	✗	0.342	0.545	0.543	0.118	0.602	81983747	42.8
SparseBEV [48]	ResNet50	✗	0.265	0.404	0.657	0.624	1.505	40657321	31.9
HoP [49]	ResNet50	✓	0.370	0.574	0.569	0.116	0.419	86832001	28.6
RailBEV (ours)	ResNet50	✗	0.385	0.573	0.492	0.113	0.588	74801868	37.6

B. Comparison With Different 3-D Object Detection Networks

To evaluate the method for railway 3-D object detection, RailBEV is compared with existing state-of-the-art 3-D object detectors, including BEVDet, BEVDet4D, BEVFormer, BEVDepth, PETR, PETRv2, SparseBEV [48], and HoP [49]. Considering the limited applicability of current BEV data augmentation strategies, no data augmentation methods are employed for all networks. Furthermore, because accurate global positions cannot be obtained during train operation, the data-collecting train is assumed to remain stationary for some BEV networks that require temporal historical frames and temporal global position. The number of temporal frames is defaulted to a 5-frame sequence.

The mmdetection3d toolkit is used for training and testing. For all networks, the Adam optimizer was utilized, and the learning rate was set to $2e^{-4}$ with a batch size of 16 per GPU.

ResNet50 and VOVNet-99 were utilized as backbones based on the model. Furthermore, BEVDet4D, BEVformer, and PETRv2 employ five historical frames as temporal inputs, while HoP uses one historical frame as additional input. All networks were trained for 24 epochs and the loss has converged.

Table II presents the 3-D detection performance for the testing set of railway detection datasets for the proposed RailBEV method and other state-of-the-art methods. Without temporal data, RailBEV significantly enhances the effectiveness of 3-D railway object detection, achieving 38.5% mAP and 57.3% NDS, outperforming the baseline model BEVDepth by 4.3% and 2.8%, respectively. Compared with HoP, the current SOTA 3-D detection method in nuScenes, RailBEV, demonstrates a clear improvement in mAP (1.5%) with a marginal 0.1% NDS reduction. Furthermore, Table II can reveal that networks using temporal data, such as BEVFormer, BEVDet4D, and PETRv2, do not significantly increase accuracy due to the absence of precise temporal ground truth. In contrast, depth-supervised networks such as BEVDepth, HoP, and RailBEV exhibit notable precision improvements. Although achieving superior results, RailBEV maintains a compatible or even lower computational cost compared to the baseline methods, which is vital for edge computing devices such as Xavier and Orin.

Fig. 9 shows the capability of RailBEV to detect objects of multiple sizes and distant objects in a variety of scenarios. The

TABLE III
EXPERIMENTS USING THE TWO PROPOSED MODULES

Model	mAP	NDS	mATE	mASE	mAOE
Original BEVDepth	0.342	0.545	0.543	0.118	0.602
+DSVT module	0.369	0.559	0.519	0.118	0.619
+LRS module	0.359	0.545	0.529	0.118	0.690
RailBEV	0.385	0.573	0.492	0.113	0.588

first row illustrates low-light scenes with severe light pollution. The second row shows detection results under normal conditions. The third row demonstrates multiscale object detection performance. Leveraging depth-supervised BEV perspective transformation, RailBEV adapts to diverse railway scenarios, including challenging lighting conditions, multiscale objects, and distant objects.

Furthermore, the detection performance of RailBEV is evaluated across various distance intervals, each spanning 30 m. As shown in Fig. 10, RailBEV is effective for distant detection and adept at identifying severely occluded and short-range objects, showcasing its adaptability for comprehensive 3-D railway object detection.

C. Ablation Study

This section presents experiments designed to validate the efficiency of the DSVT and LRS modules. The hyperparameters and settings are the same as those for RailBEV, as described in Section IV-B. BEVDepth is used as a baseline model. Experiments verified the effectiveness of the proposed modules and the submodule in the DSVT module, including ground-truth depth labels, multiscale structure, and mixed-pooling loss.

1) *Experiments on Two Modules*: The proposed modules DSVT and LRS can be used individually to improve accuracy. Thus, experiments were conducted to test the accuracy of the two modules separately. As shown in Table III, the first row is the original BEVDepth, and the second and third rows represent the DSVT and LRS modules, respectively.

As shown in Table III, both modules improved their performance significantly. DSVT module achieved a 2.7% gain in terms of the mAP and a 2.4% decrease in the mATE with low model complexity. This indicates that the DSVT module enables accurate depth estimation of BEV features. Furthermore, the LRS module also improved the results by

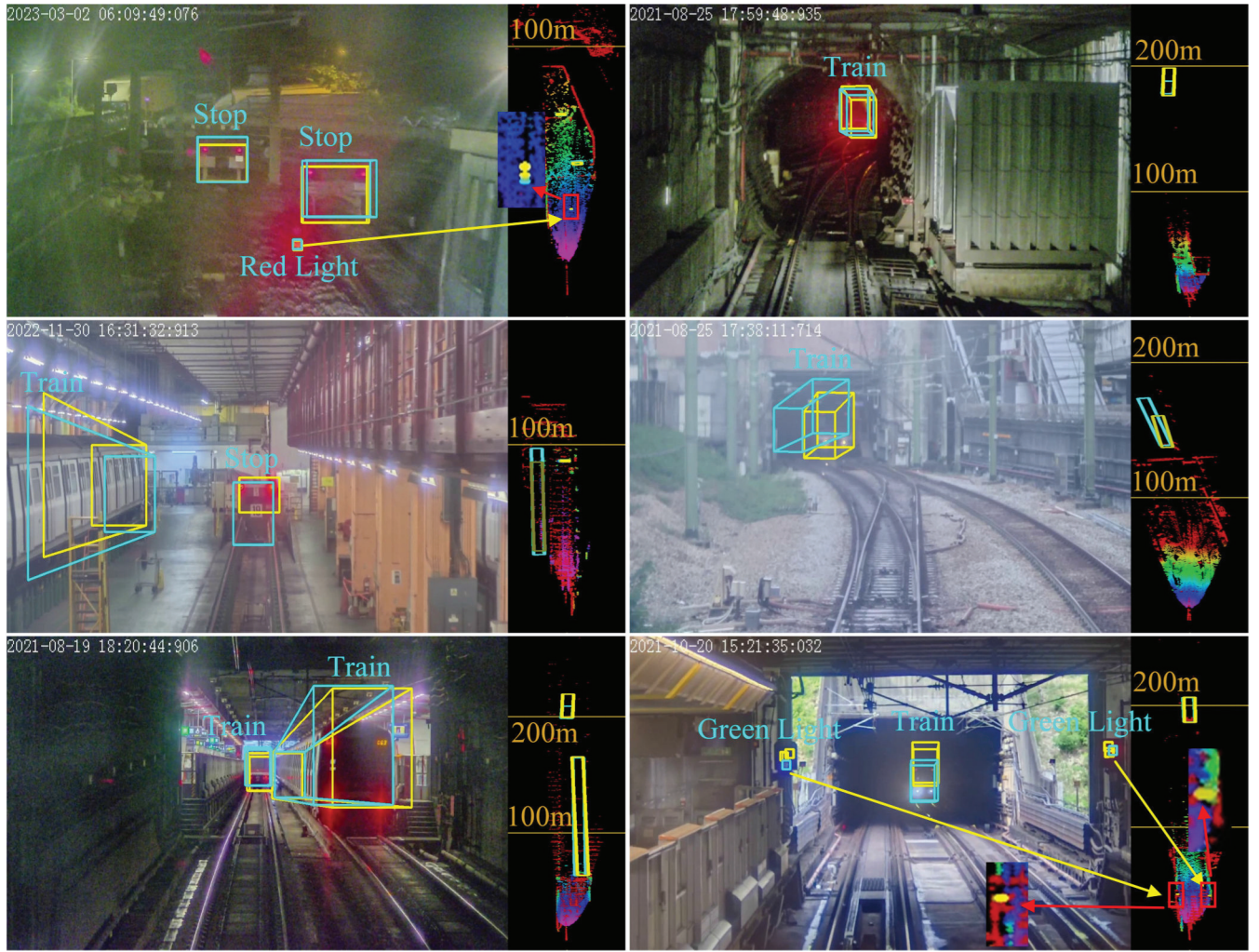


Fig. 9. Experimental results for the proposed RailBEV. The blue boxes refer to the ground truth of 3-D objects, whereas the yellow boxes represent the predicted bounding boxes. The area of signal lights is magnified, and their position is highlighted with a red arrow due to their small size. Otherwise, they would be difficult to observe in BEV images. Additionally, yellow horizontal lines serve as distance markers at 100- and 200-m intervals within the BEV image.

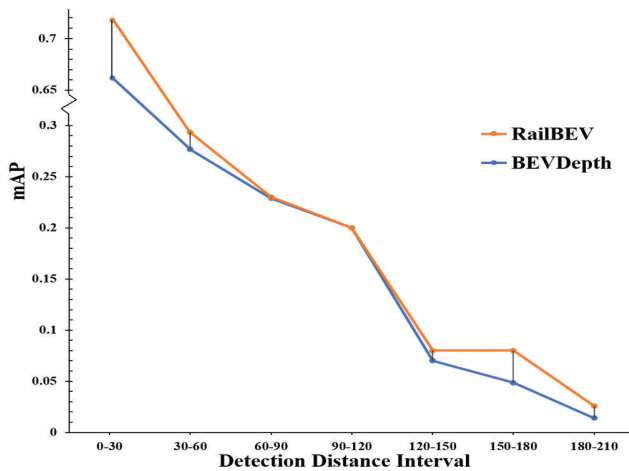


Fig. 10. Detection accuracy of RailBEV and BEVDepth at different distance intervals. The length of each interval is 30 m.

over 1.7% compared to the baseline with a low computational cost. This implies that the LRS module can not only effectively

learn long-range 3-D knowledge but is also suitable for short-range objects.

2) *Experiments on DSVT Module:* The DSVT module described in Section III-A contains the depth image generated, multiscale network, and mixed-pooling loss. To investigate the effectiveness of the submodule, the view transformation method of the original BEVDepth is replaced with the DSVT module, which means that a dense depth image multiscale structure is used and the mixed-pooling loss is employed.

As shown in Table IV, the three submodules improved the performance significantly. First, a multiscale depth supervision structure is tested, which achieves a 1.2% improvement with low computation cost. This indicates that the multiscale depth supervision method can learn more depth estimation information than the original method. Subsequently, dense depth images and mixed-pooling loss are added, respectively, improving the results by 1.6% and 2.1% compared to the original methods. Notably, the utilization of dense depth images significantly improves the NDS, confirming the significant role of dense depth images in the estimation of object positioning.

TABLE IV

EXPERIMENTS WITH DSVT MODULE. “DI,” “MS,” AND “MPL” DENOTE THE DENSE DEPTH IMAGE, MULTISCALE NETWORK, AND MIXED-POOLING LOSS

MS	DI	MPL	mAP	NDS
			0.342	0.545
✓			0.354	0.555
✓	✓		0.363	0.560
✓		✓	0.358	0.555
✓	✓	✓	0.369	0.559

TABLE V

EXPERIMENTS WITH DIFFERENT DEPTH IMAGE RESOLUTION

Resolution	mAP	NDS	FPS
88×160	0.384	0.563	15.4
44×80	0.385	0.573	37.6
22×40	0.374	0.566	42.7
11×20	0.365	0.552	44.3

3) *Experiments on Different Depth Image Resolution:* In the DSVT module, depth images of varying scales are unified to a resolution of 44×80 for subsequent BEV perspective transformation. To assess the impact of depth map resolution on accuracy and computational efficiency, resolutions of 88×160 , 44×80 , 22×40 , and 11×20 were evaluated.

As shown in Table V, the 44×80 resolution selected for RailBEV achieves optimal mAP and NDS. It outperforms the 88×160 resolution in NDS by 1% while improving processing speed by 22.2 FPS. Lower resolutions, such as 22×40 , also yield competitive results, with a marginal decrease in mAP and NDS but a further 5.1 FPS increase, highlighting a tradeoff between accuracy and computational efficiency.

V. CONCLUSION

By focusing on 3-D railway object detection, this study proposes a vision-based BEV-representation network called RailBEV. This network is divided into two parallel modules: the DSVT module and the LRS module. The DSVT module generates dense depth images, utilizing a multiscale depth supervision network to enhance spatial depth knowledge during the view transformation process. Finally, a mixed-pooling loss is proposed to facilitate depth supervision. Furthermore, the LRS module predicts the 3-D location heatmap from 2-D image features, guiding the image backbone to learn 3-D information and enhancing the BEV features with depth-aware attention. In addition, an anchor-free detection head called CenterHead is employed for 3-D object detection. The experimental results show that RailBEV has achieved desirable results of 38.5% and 57.3% in terms of mAP and NDS, respectively. Notably, RailBEV improved the accuracy of distant object detection. Within each distance interval, the method demonstrates superior precision compared with the baseline approach.

However, due to the challenging light conditions in railway tunnels, which significantly affect images, numerous false detections have been observed in certain scenarios. Moreover,

because of the limitations of depth estimation, the accuracy of distant object location prediction is undesirable. In future work, it is expected to fuse images with LiDAR point clouds to achieve a denser representation and capture more robust environmental data on railways.

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